

# Jorge Manuel Torre

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## Education

### Virginia Tech

B.S. Computer Science, Overall GPA: 3.32/4.0, Major GPA: 3.50/4.0

Blacksburg, VA

May 2026

**Relevant Coursework:** Security of AI, Machine Learning (CS 4824), Linux Kernel Programming, Software Reverse Engineering, Computer Architecture, Computer Systems, Data Structures & Algorithms.

## Projects & Experience

### CHUD: Catastrophic Harmful Update Detection • GitHub

January 2026 – April 2026

*Python, PyTorch, LoRA, Llama-2-7b, Llama-Guard-3-8b, Unsloth*

*Team of 4 (CS Capstone)*

- Tested the LoX defense (Perin et al., COLM 2025) against an iterative LoRA fine-tuning attack on Llama-2-7b using benign GSM8k and adversarial BeaverTails samples.
- Built an Attack Success Rate (ASR) evaluation pipeline with Llama-Guard-3-8b as the judge model, calibrated it against AdvBench (99% accuracy, 2% FNR), and ran 30+ experiments.
- Found that LoX lowered ASR from 91–96% to 37–63% under direct harmful fine-tuning, but an iterative mixed-data attack raised ASR back to 89%.
- Set up x86-64 and aarch64 VT GPU cluster environments for the team and ran QLoRA fine-tuning evaluations for the final analysis.

### Celeste-RL: Deep Reinforcement Learning for a Precision Platformer • GitHub • Website

January 2026 – April 2026

*Python, PyTorch, NumPy, matplotlib, CUDA*

- Compared five RL methods (DQN, Behavioral Cloning, Hybrid, Curriculum, Dueling DQN with curiosity) trained to play Celeste Classic room 0 on the Pyleste emulator.
- Found that state representation mattered more than algorithm choice: replacing raw PICO-8 tile IDs with a 5-class encoding raised plain DQN completion from 8% to 57–68% with no other change.
- Built scripts for batched evaluation, comparison plots, side-by-side gameplay GIFs, a GitHub Pages results site, and a debugging devlog covering 9 iterations.

### Garage64: Custom Entity Model Studio • GitHub

August 2025 – Present

*React, Three.js, Django, SQLite, Docker*

- Developing a self-hosted web application for creating custom Minecraft vehicle models and texture packs using the OptiFine Custom Entity Model (CEM) format.
- Building an interactive React and Three.js 3D editor that supports real-time vehicle customization through modular part swapping and texture painting.
- Implementing a Django REST API for managing entity metadata, asset configurations, and project persistence.
- Creating an automated export pipeline for generating .jem and .jpm asset files from user-created models.
- Containerizing the application with Docker Compose to streamline local development and deployment.

### Experimental Home Lab

January 2021 – Present

*Proxmox, OPNsense, NixOS, Docker, CUDA*

- Run a multi-VM Proxmox lab with NixOS / Docker workspaces and a CUDA-enabled training environment used for the ML projects above.

## Leadership & Community

### Team Lead, Database Management Systems (CS 4604)

May 2025 – August 2025

- Led a 3-person team through schema design and SQL development, then integrated the team's work into the final submission.

### Team Lead, Intermediate Software Design (CS 3704)

January 2025 – April 2025

- Led a 5-person team through requirements gathering, sprint planning, code reviews, and final project delivery.

### Garage32 (Community Project)

March 2020 – December 2021

- Led development and distribution of a graphical asset pack for the Minecraft community, reaching **600+ downloads**; promotional content earned **30K+ likes** on a single TikTok post.

## Skills

**ML / AI:** PyTorch, LoRA, Fine-Tuning, LLM Safety & Alignment, Reinforcement Learning (DQN, BC, Curriculum), Llama / Llama-Guard, Adversarial ML, Research Methodology

**Languages:** Python, C, C++, Java, SQL, JavaScript, Assembly (ARM, x86-64, RISC-V), Bash

**Infrastructure & Tooling:** CUDA, Docker, NixOS, Proxmox, Linux, Git, CI/CD

**Also familiar with:** React, Three.js, Django, PostgreSQL